

Programming Exam 5

Date/Time: 2026.04.21 08:00 – 08:50

(程式檔命名學號_pe5.py，上傳至 Moodle PE5 上傳區)

Problem: Mine Hint Board

Your task is to simulate a simple **mine hint board** like the classic Minesweeper game. The board has multiple rows and columns. Some cells contain mines, and all other cells should show the number of mines in their surrounding neighboring cells. A neighboring cell can be horizontal, vertical, or diagonal.

Write a Python program that:

- Take the dimensions of the board as input (number of rows and columns).
- Take the mine locations as input in the format `index|index|index|...`. Treat each mine location as a **single linear index**, starting from 0, counting from left to right and top to bottom.
- Check whether each mine index is **out of range**. If an index is smaller than 0 or greater than or equal to the total number of cells, print `Invalid index: x`, where x is the invalid index.
- Ignore invalid indices and only places valid mines on the board.
- Convert each valid linear index into its corresponding row and column position.
- Build the board using a **nested list**.
- Place `*` for mine cells. Fill every non-mine cell with the number of adjacent mines.
- Display the final board as strings, with each row on a new line and cells separated by a whitespace.

Sample Input/Output

(以下是你程式執行後須印出的結果)(繳交是交 pe5.py 檔，不是交截圖)

```
c:\workspace>python pe5.py
4 4
1|6|8|13
1 * 2 1
2 3 * 1
* 3 2 1
2 * 1 0

c:\workspace>python pe5.py
3 5
0|4|7|14
* 2 1 2 *
1 2 * 3 2
0 1 1 2 *

c:\workspace>python pe5.py
3 4
2|5|12|-1|9
Invalid index: 12
Invalid index: -1
1 2 * 1
2 * 3 1
2 * 2 0

c:\workspace>
```

Note: You need to write comments (註解) for each part in your code.